Torsion

Objective

Determination of angle of twist and performing theoretical and experimental data validation of torsion equation.

Apparatus

The apparatus (Fig.1) setup consists of work panels for holding the specimen and is provided with a weight hangers and weighing pans for loading the shaft in pure torsion. A rotating chuck with angular measurements are used to measure the twist of the shaft.

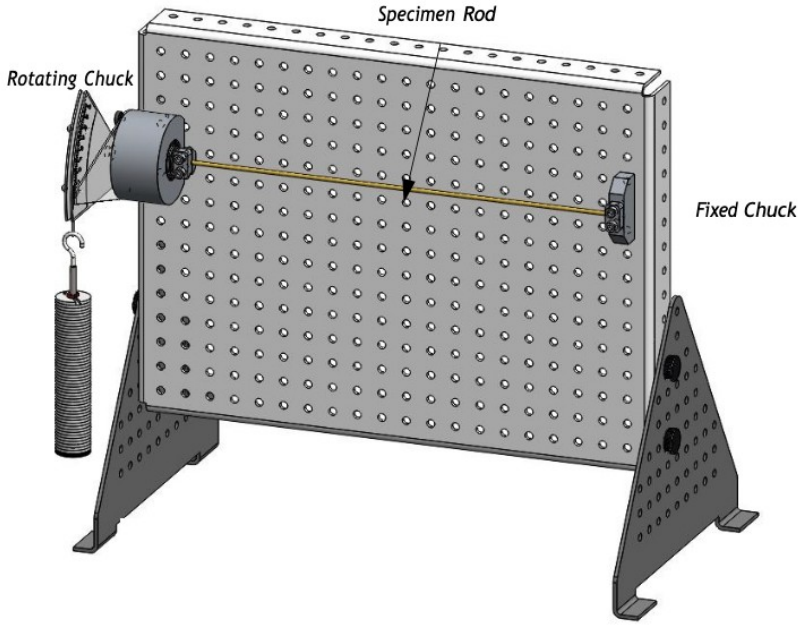


Fig .1 Experimental setup

Theory

A slender member subjected primarily to twist is usually called a shaft. Shafts are used in the transfer of mechanical power from one point two another. In such an application, one is primarily interested in the twisting moment, which can be transmitted by the shaft without damage to the material. Knowledge of stresses that develop due to twisting (and its variation in the cross section) is necessary to be known. In certain applications twisted shafts are act as a spring with prescribed stiffness with respect to rotation. In such a case, one I interested primarily in the relation between the applied twisting moment and the resulting angular twist of the shaft. For a circular shaft of constant diameter transmitting a uniform torque

$$\frac{M}{J} = \frac{G\varnothing}{L} = \frac{\tau}{r}$$

Where M is the twisting moment applied, J is the polar moment of inertia of the cross-section τ is the shear stress developed due to torsion, r is the radius of the element considered, G is the shear modulus of the material, ∅ is angle of twist and L is the length of the uniform shaft.

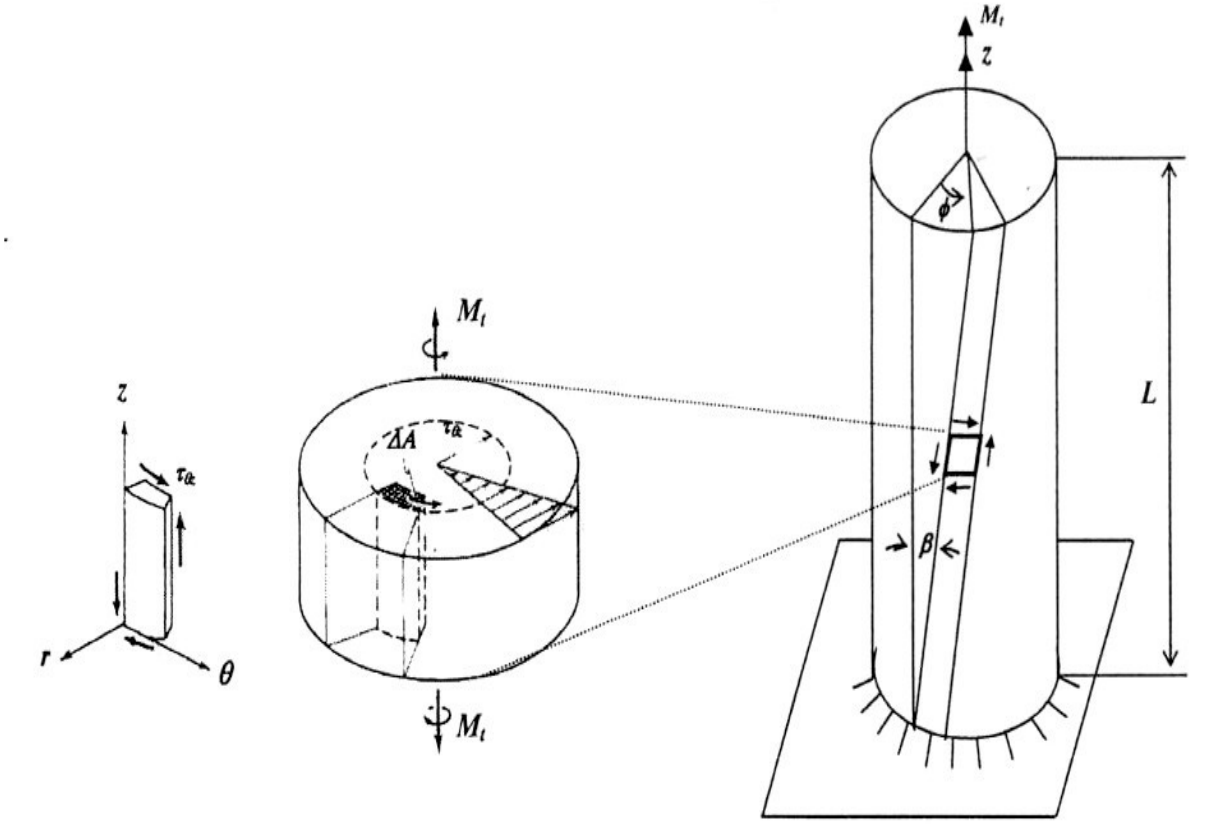


Fig. 2 Figure showing shear stress distribution

Fig. 3 Figure showing the definition of angle of twist

Procedure

1) Measure the cross sectional details of the shaft.
2) Fix the shaft in the setup
3) Adjust the rotating chuck in such a way that after hanging the weight hanger, reading is set to zero.

4) Measure the length of the shaft (L)
5) In increments of 100 g, add weights up to a maximum of 500 g to the cord. Each time you add a weight, tap the Work Panel to reduce the effects of friction. Record the twist angle at the Moment Arm at each increment.
6) Figure out how pure torsion is applied to the shaft.
7) Due to loading the shaft is twisted. Note down the corresponding reading.
8) Figure out how to find out the angle of twist from these readings.
9) Repeat the same for brass, steel, & aluminium samples

Analysis of Results

1) Draw the free body diagram of the loaded specimen.
2) Plot a graph showing the torque versus angle of twist

Table

shaft dimensions r = L=

Load(m) (g)	Load(W) (N)	Torque (T) (Nm)	Actual Twist Angle (degrees)	Actual Twist Angle (radians)	Theoretical Twist Angle (degrees)	Theoretical Twist Angle (radians)
100						
200						
300						
400						
500						
Material - Brass			**G = 38Gpa**			

Load(m) (g)	Load(W) (N)	Torque (T) (Nm)	Actual Twist Angle (degrees)	Actual Twist Angle (radians)	Theoretical Twist Angle (degrees)	Theoretical Twist Angle (radians)
100						
200						
300						
400						
500						
Material - Steel			**G = 79.6 Gpa**			

Load(m) (g)	Load(W) (N)	Torque (T) (Nm)	Actual Twist Angle (degrees)	Actual Twist Angle (radians)	Theoretical Twist Angle (degrees)	Theoretical Twist Angle (radians)
100						
200						
300						
400						
500						
Material - Aluminium			**G = 25.5 Gpa**			

Discussion

1) How is shear Modulus related to Young's modulus?

2) Look at the general torsion equation and confirm that the values of the angle of twist and torque are constant
   Do your results clearly show the relationship between torque and angle of twist?

3) Does the experiment clearly show the relationship between the length of the rod and angle of twist (with a constant load). Comment on any possible causes of error.